Forensic Knowledge Graphs as investigative infrastructure: executive summary

Tai Nguyen · October 28, 2026

For Jane and her student. A short view of the proposal, the evidence and the decisions the paper needs. The companion full report (`report.pdf`) has each study's setup, per-question tables, robustness and agent results, and the code. All results come from synthetic worlds with planted ground truth.

The proposal

Treat each FKG record (arXiv:2606.23917) as linked data that shares identifiers with how the image was posted, what was claimed about it, which credentials it carries and which detector produced the analysis. Investigative questions that neither forensic analysis nor platform context can answer alone then become standard graph queries, written by an analyst or by an LLM agent.

Novelty in one sentence: forensic observations become first-class linked data, so an investigation is a composition of standard queries over shared identifiers rather than a bespoke pipeline per question.

	Contribution	What it is
C1	**Encoding**	FKG in OWL 2 and RDF with a SHACL profile, plus a thin context layer that reuses Activity Streams 2.0, schema.org Claim and ClaimReview, C2PA and IPTC, and PROV-O. Named graphs record who asserted what.
C2	**Expressiveness**	18 investigative use cases in four families (time, copies, claims and intent, auditing the analysis), each a competency question with a runnable SPARQL query.
C3	**Evidence**	Four case studies with planted ground truth, one followed over a simulated year of campaign waves. The integrated query beats each single layer and degrades gracefully up to 50% noise.
C4	**Agentic investigation**	LLM agents given only the schema and a page of notes write queries that match or beat hand-written ones.

Headline results

- **The gain comes from integration, not detection.** Only the joins differ between arms. Forensics alone cannot see time, claims or accounts; context alone cannot see what was altered.
- **Errors are misses, not false accusations.** Noise lowers recall while precision holds: 1.00 at every noise rate for CS3 triage, at least 0.81 for CS1 recycled-media flags up to $\eta = 0.5$. At $\eta = 0.5$, CS1 macro-F1 is 0.786 against 0.594 for forensics alone.
- **Defence is a new query, not a new model.** In CS4, each attack that broke detection was answered by a new query over the same records: of seventeen attacks, seven fixed, four partly, six open.
- **Forensic TTPs outlast personas.** When operators retire every account after each wave, the processing-chain join links 85% of new waves to the right operator and none to the wrong one;

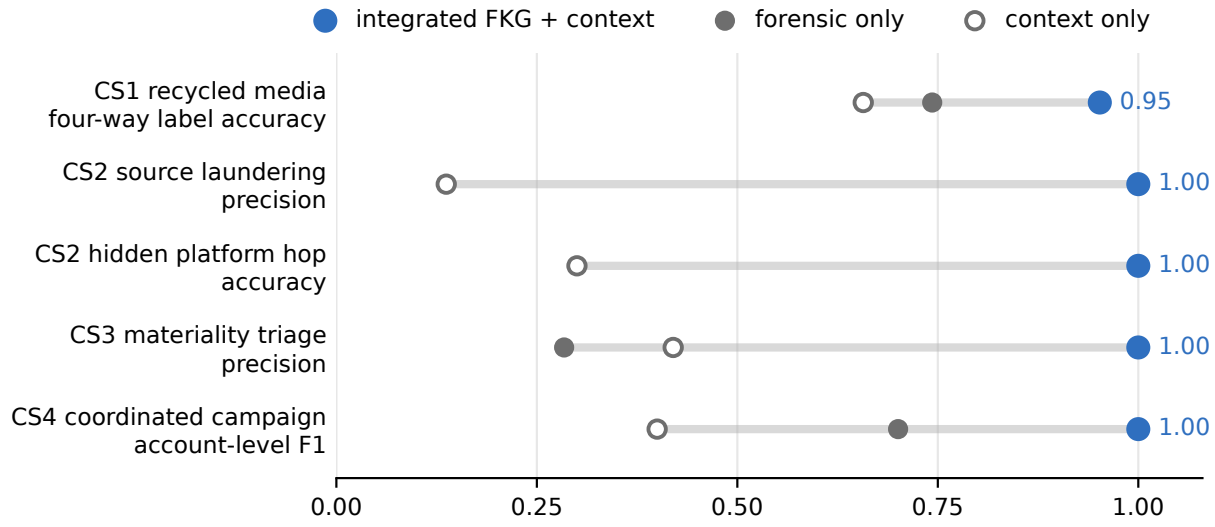


Figure 1: Headline results, one row per investigative question. Every arm reads the same detector outputs.

account overlap links none, and narrative or timing name the wrong operator in 12 to 28% of waves. It holds over a year at ten times the population (4.84M quads, 19 of 22 waves) and among 23 benign look-alike accounts per campaign account, where within-wave detection precision falls from 0.87 to 0.41 (Figure 3).

- **The schema is enough for agents.** On held-out worlds, agent-written queries score 0.92 to 0.96 against 0.91 for the hand-written set.

How it works

- **Three identifiers do the joining:** the image IRI (post, credential, FKG record), the entity IRI (claim subject and altered content) and the detector run (FKG record and the fact-check that cites it).
- **Almost nothing is new vocabulary.** The forensic layer is FKG as published; the context layer borrows existing standards plus a handful of terms such as a platform's transcoding profile.
- **Plain SPARQL 1.1, no reasoner.** SHACL validates every record.

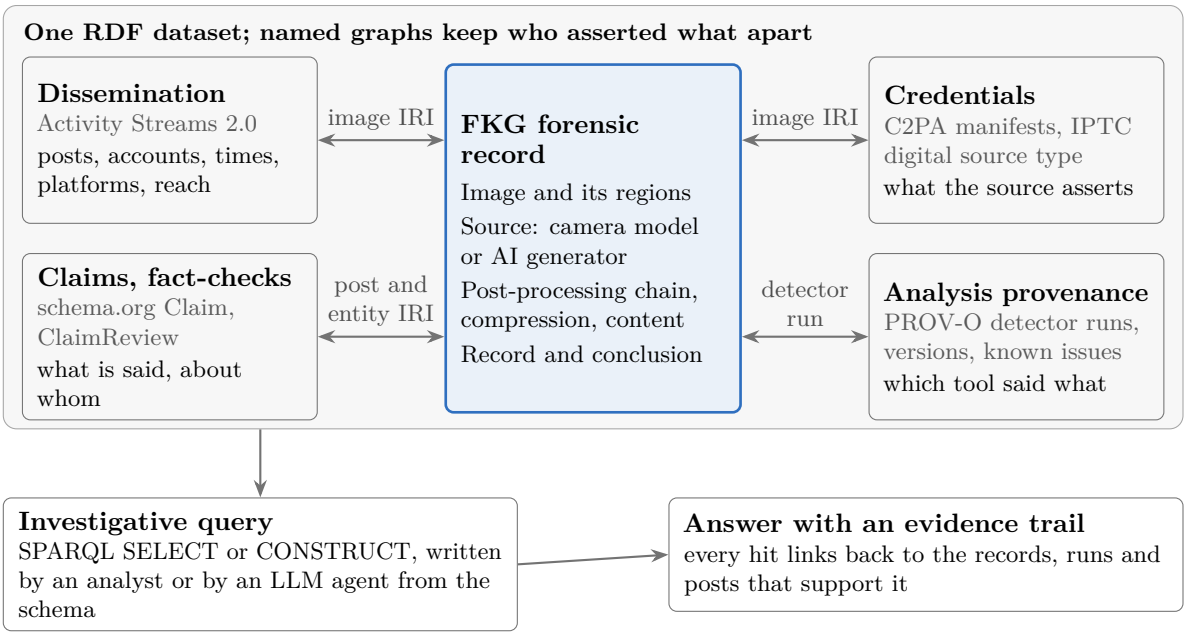


Figure 2: Architecture. The FKG record is unchanged; everything around it reuses an existing standard. Labels on the arrows are the join keys.

Four case studies at a glance

Study	Investigative question	What the join adds	Integrated	Best single layer
CS1 Recycled media	Is an authentic photo presented as showing an event it predates?	Earliest near-duplicate copy against the claimed event date, guarded by camera-model agreement	0.952 accuracy	0.743 (forensic)
CS2 Lineage and laundering	Which copy added the AI region, which platform did a file pass through, who credits an outlet for an edit it never published?	Copies ordered by compression, resampling and inserted regions, then tied to posts and credits	Hop 50/50, laundering precision 1.00	15/50, 0.137 (context)
CS3 Materiality and intent	Does an undisclosed alteration touch what the claim is about?	Altered content's entity against the claim's subject and the credential	Precision 1.00 at recall 0.912	0.420 (context)
CS4 Coordinated campaign	Which accounts run a coordinated campaign with manipulated media?	Shared generator and processing recipe (a forensic TTP) plus narrative and timing	F1 1.00 (0.97 at $\eta = 0.7$)	0.70 (forensic)
CS4 across waves	Does a wave of fresh accounts come from a known operator?	The same TTP matched to accounts that lagged takedown reports name, chained through earlier waves	85% of waves right, none wrong	narrative, 12% wrong

The graph also audits its own analysis (CS3). It finds all 8 credentials the forensics contradict, all

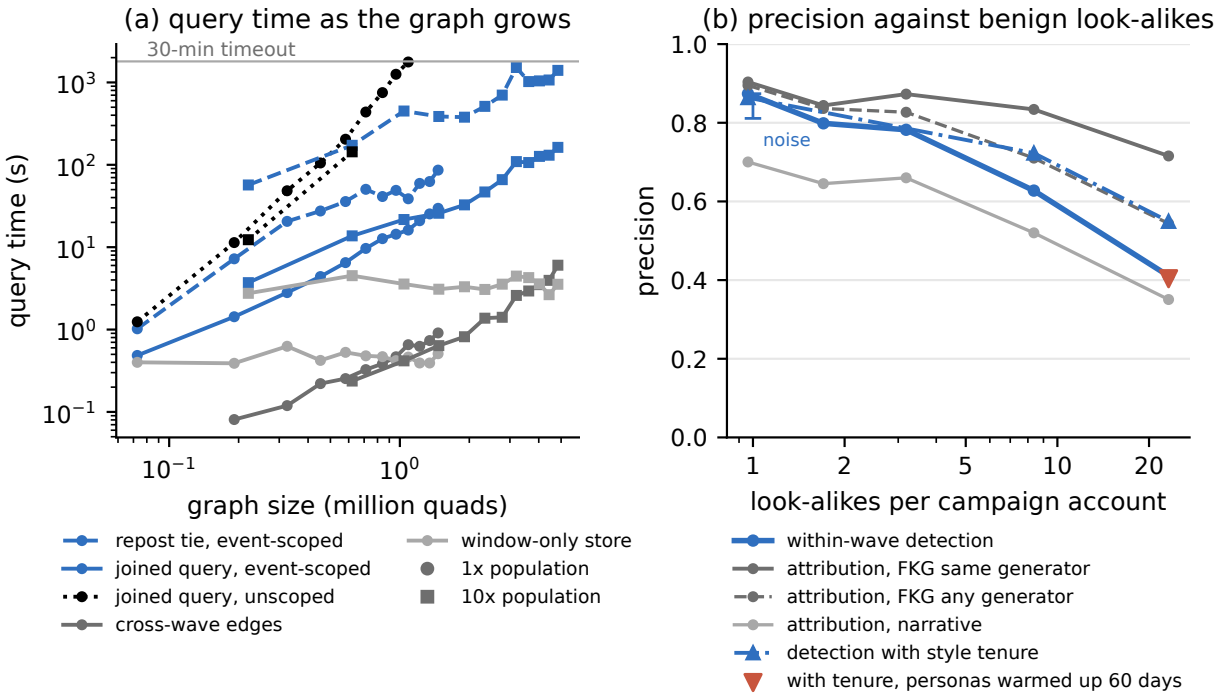


Figure 3: CS4 over a simulated year. (a) Query time against graph size at 1x and 10x population; scoping to the event keeps the joined query linear. (b) Precision against benign look-alike accounts per campaign account: detection precision halves while same-generator attribution holds best.

3 attributions to a generator not yet released, and all 5 fact-checks that rest on a faulty detector version.

Decisions the paper needs from Jane

CS1 to CS3 and CS4 were built in parallel and made different, reversible encoding choices. The six that matter most:

Decision	Question	Our suggestion
Where new terms live	Should the context vocabulary join FKG 0.2, or ship as a profile that imports it?	Open
Fully synthetic images	A verdict value, a manipulation type, or both?	One shared term
Named graphs	Per forensic record, per producer, or both?	Per producer
Statement-level provenance	Named graphs, or RDF 1.2 triple terms?	Open
Image identifiers	Adopt content-derived `ni:///sha-256` IRIs?	Adopt
Entities on content	Should Content carry entity IRIs natively? CS3 depends on it.	Adopt

Four smaller ones (compression subclasses, axioms SHACL cannot check, a shared reliability threshold, where an intent assessment belongs) are in §8 of the full report.

Caveats

- **Synthetic only.** The numbers show that queries recover planted structure, not how an FKG pipeline performs on real media. The paper's evaluation should open with this.
- **Circularity risk.** One author built the worlds and the hand-written queries. Agents that never saw those queries, fresh seeds and held-out worlds mitigate it.
- **Leaks found and fixed.** Both simulators first leaked ground truth into readable attributes. After the fix, no single attribute predicts truth above about 0.67 balanced accuracy.
- **Upper bounds by design.** A pasted region's single-compression trace is recorded without noise, and CS4's processing-chain detection rates are assumed.
- **Hardest question.** CS2 patient zero reaches precision 0.75 (F1 0.56 across ten seeds): a detector false positive that recurs on copies looks like corroboration.
- **Scale.** Worlds hold 10k to 4.84M quads. Scoping each query to its event keeps the joined query at seconds to minutes where the unscoped one passes 30 minutes; real crawls still need incremental candidates and materialised rules.
- **Base rates.** Benign coordination (fans, brand pushes, news syndication) halves within-wave precision at 23 look-alikes per campaign account, and a house-style tenure filter recovers it only until operators warm personas up for 60 days. The real ratio is unknown and should be measured before any precision claim.

Next experiments and code

Experiment	What it would establish	Effort
Real FKG pipeline on public out-of-context and provenance benchmarks	The joins still help with real detector outputs	Medium
Join real published fact-checks to the images they cite	The claim layer works on messy real claims	Medium
Settle the decisions; release FKG 0.2, the profile, shapes and 18 competency questions	A reusable benchmark for query-based investigation	Low
Controlled agent study: several models, repeated runs, notes with and without caveats	How reliably agents produce correct, affordable queries	Medium
Measure benign fan, brand and syndication clusters on a real platform sample	The precision a deployed campaign detector can expect	Medium

Code: private GitHub repository ductai199x/fkg-case-studies, managed with uv. After `uv sync`, `make all` rebuilds every world and result, and `make check` audits every SPARQL query and reruns the quoted numbers; `make sweep` and `make agents` run separately. §11 of the full report lists one command per result.